Experiment-9 : ANALYSIS OF PHOTODETECTOR NOISE AND SNR USING MATLAB/SCILAB

AIM:

To analyse photodetector noise and calculate the Signal-to-Noise Ratio (SNR) using MATLAB/SCILAB.

OBJECTIVES:

- To study shot noise and thermal noise.
- To calculate total noise current.
- To determine the Signal-to-Noise Ratio (SNR).
- To plot noise and SNR graphs.

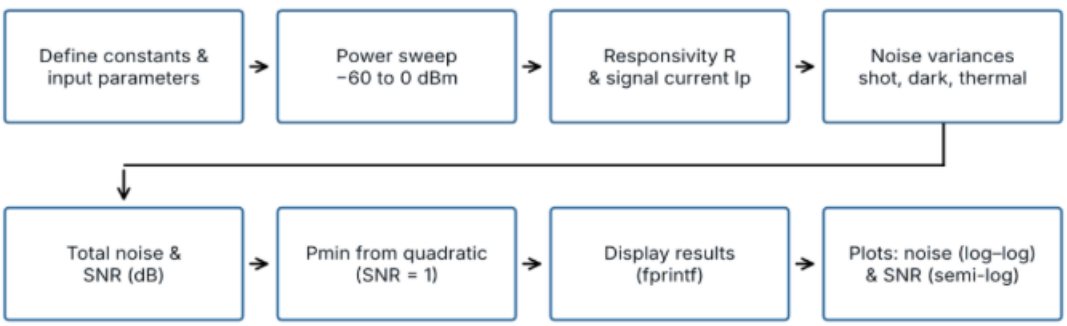
SOFTWARE REQUIRED:

- MATLAB
- LAPTOP

EXPERIMENT PROCEDURE:

- Open MATLAB/SCILAB software.
- Create a new script and enter the program.
- Define the physical constants and photodetector parameters.
- Calculate shot noise and thermal noise.
- Determine the total noise current.
- Calculate the SNR value.
- Plot the noise and SNR graphs.
- Observe and analyse the output graphs.

BLOCK DIAGRAM:

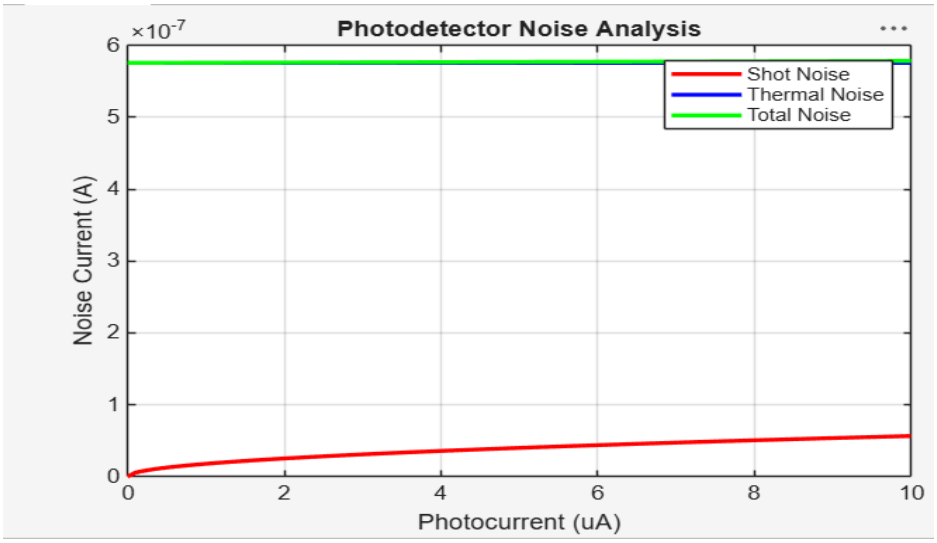


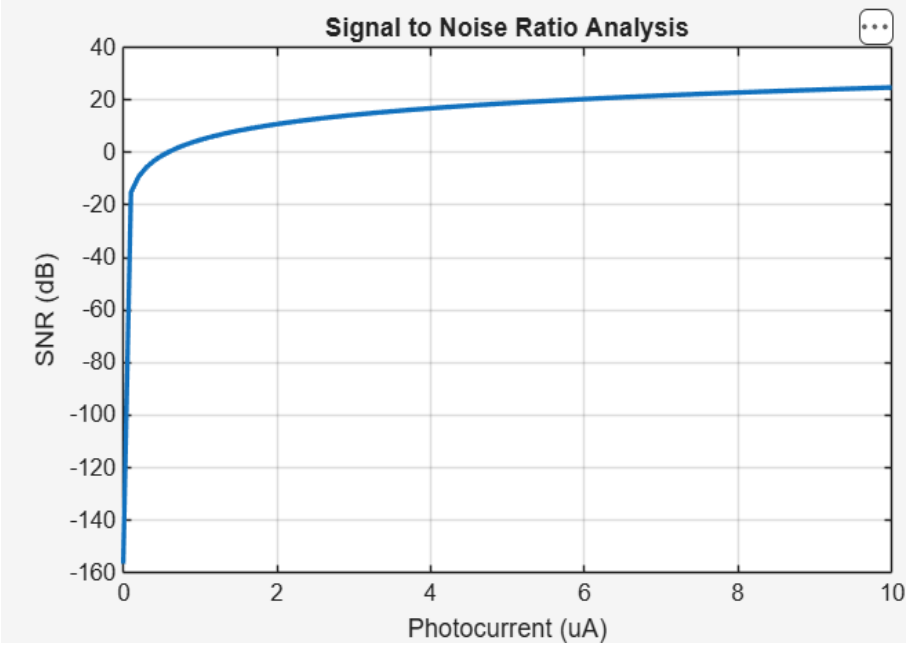
PROGRAM:

```
clc;
clear;
close all;
q = 1.6e-19;
Id = 1e-6;
B = 1e9;
R = 50;
T = 300;
k = 1.38e-23;
I = 0:0.1e-6:10e-6;
shot_noise = sqrt(2*q*I*B);
thermal_noise = sqrt((4*k*T*B)/R);
total_noise = sqrt(shot_noise.^2 + thermal_noise^2);
signal_power = (I.^2)*R;
noise_power = total_noise.^2*R;
SNR = signal_power ./ noise_power;
SNR_dB = 10*log10(SNR + eps);
figure;
plot(I*1e6,shot_noise,'r','LineWidth',2);
hold on;
plot(I*1e6,thermal_noise*ones(size(I)),'b','LineWidth',2);
plot(I*1e6,total_noise,'g','LineWidth',2);
xlabel('Photocurrent (uA)');
ylabel('Noise Current (A)');
title('Photodetector Noise Analysis');
legend('Shot Noise','Thermal Noise','Total Noise');
grid on;
figure;
plot(I*1e6,SNR_dB,'LineWidth',2);
```

```
xlabel('Photocurrent (uA)');
ylabel('SNR (dB)');
title('Signal to Noise Ratio Analysis');
grid on;
disp('Photodetector Noise and SNR Analysis');
disp(['Thermal Noise = ',num2str(thermal_noise),' A']);
disp(['Maximum SNR = ',num2str(max(SNR_dB)),' dB']);
```

OUTPUT:





REFLECTION:

- Studied shot noise and thermal noise in photodetectors.
- Understood the relationship between photocurrent and SNR.
- Learned to analyse noise using MATLAB/SCILAB.

RUBRICS:

The experiment helped to understand the performance of an optical communication link using OptiSystem. The eye diagram was used to observe signal quality, noise and intersymbol interference

Assessment Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Weightage
System Design	Correct optical link design in OptiSystem.	Minor connection errors.	Partial system setup.	Incorrect setup.	20%
Parameter Configuration	Correct transmitter, fiber and receiver parameters.	Minor parameter errors.	Some parameters missing.	Incorrect configuration.	20%
Measurement & Monitoring	Accurate eye diagram, BER and Q-factor readings.	Minor missing readings.	Partial measurements.	No proper measurements.	20%
Signal Quality Analysis	Correct analysis of eye opening, noise and ISI.	Basic signal analysis.	Limited analysis.	No proper analysis.	20%
Report & Conclusion	Complete report with eye diagram, graphs and results.	Good report and conclusion.	Incomplete report.	Poor documentation.	20%

RESULT:

Thus, the optical fiber communication link was successfully designed and analysed using OptiSystem. Eye diagrams, BER and Q-factor were observed to evaluate the signal quality and system performance.